COMP 532
Machine Learning and Bioinspired Optimisation

Lecture 2

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Overview

- Problem solving from nature
- Intuitions from nature to AI

Why Learn from Nature?

- Natural systems have evolved highly effective mechanisms for problem solving. Despite relying on simple components and local interactions, they exhibit:
    - **Massive parallelism** – many units acting simultaneously;
    - **Emergent intelligence** – complex global behaviour arising from simple rules;
    - **Robustness and adaptability** – tolerance to noise, failure, and change.
- These properties motivate a broad class of AI methods known as **bio-inspired learning and optimisation**, which use ideas from biology to design artificial systems capable of learning and autonomous decision-making.

- We are not copying biology exactly. We abstract useful principles.

Bio-inspired Systems vs. Biological Modelling

- It is important to distinguish two related but different research directions:
- **Bio-inspired systems**
  - Use biological ideas to solve engineering problems;
  - Deliberately simplify or ignore biological details;
  - Focus on performance, scalability, and robustness.
- **Biological systems modelling**
  - Use computational systems to understand real biological processes;
  - Emphasise biological fidelity and explanatory power.

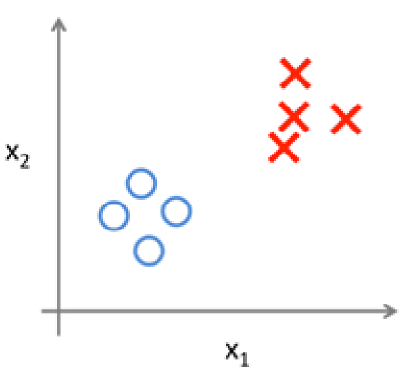
- **This module focuses on bio-inspired systems**, not biological modelling.

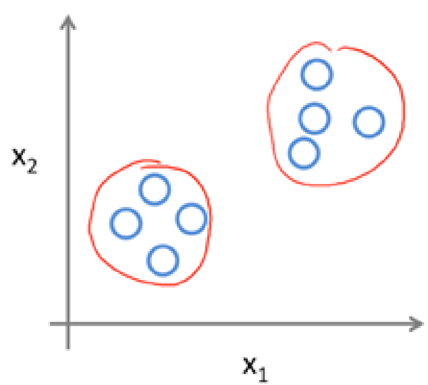
From intuition to artificial systems

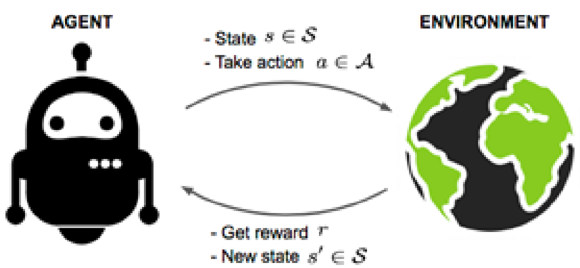
- Reinforcement Learning
- Deep Learning
- Deep Reinforcement Learning
- Multi-agent System

What is Reinforcement Learning?

Supervised Learning	Unsupervised Learning	Reinforcement Learning
• Labeled data • Direct feedback • Predict outcome/future	• No labels • No feedback • “Find the hidden structure”	• Decision process • Reward system • Learn series of actions

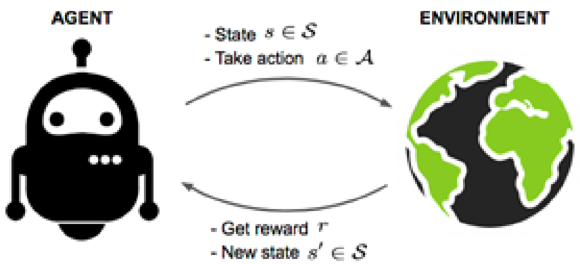






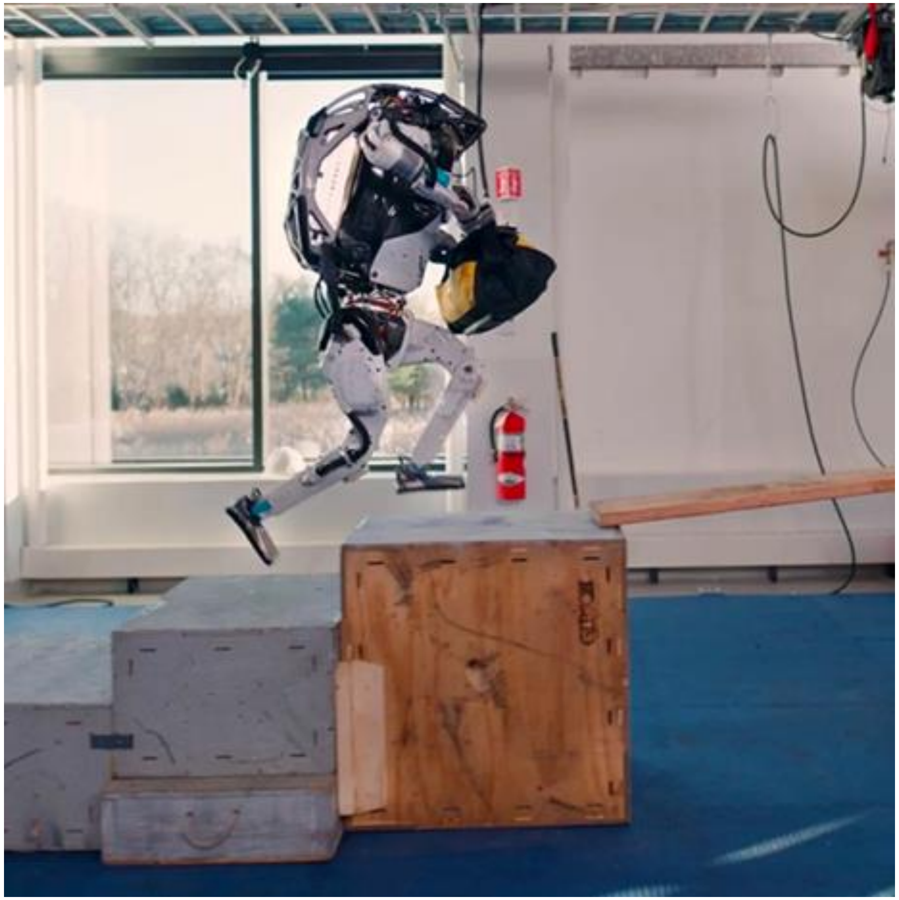
Reinforcement Learning (RL)

- Reinforcement learning studies how an **agent** learns to make decisions by interacting with an **environment**.
  - The agent observes a state;
  - Chooses an action;
  - Receives a reward;
  - Updates its behaviour to maximise long-term cumulative reward.
- Unlike supervised learning, RL does not rely on labelled data. Learning happens through **trial and error**.
- **Why is RL difficult?**
  - Rewards may be delayed;
  - Observations may be partial or noisy;
  - Environments may be unknown or stochastic.
- **Human analogy**: Learning to walk involves repeated failure, sparse feedback, and gradual improvement.



Real-world application - Robot

- Humanoid robot



RL: from intuition to artificial systems

Human vs. Agent

[Stages of Learning to Walk]

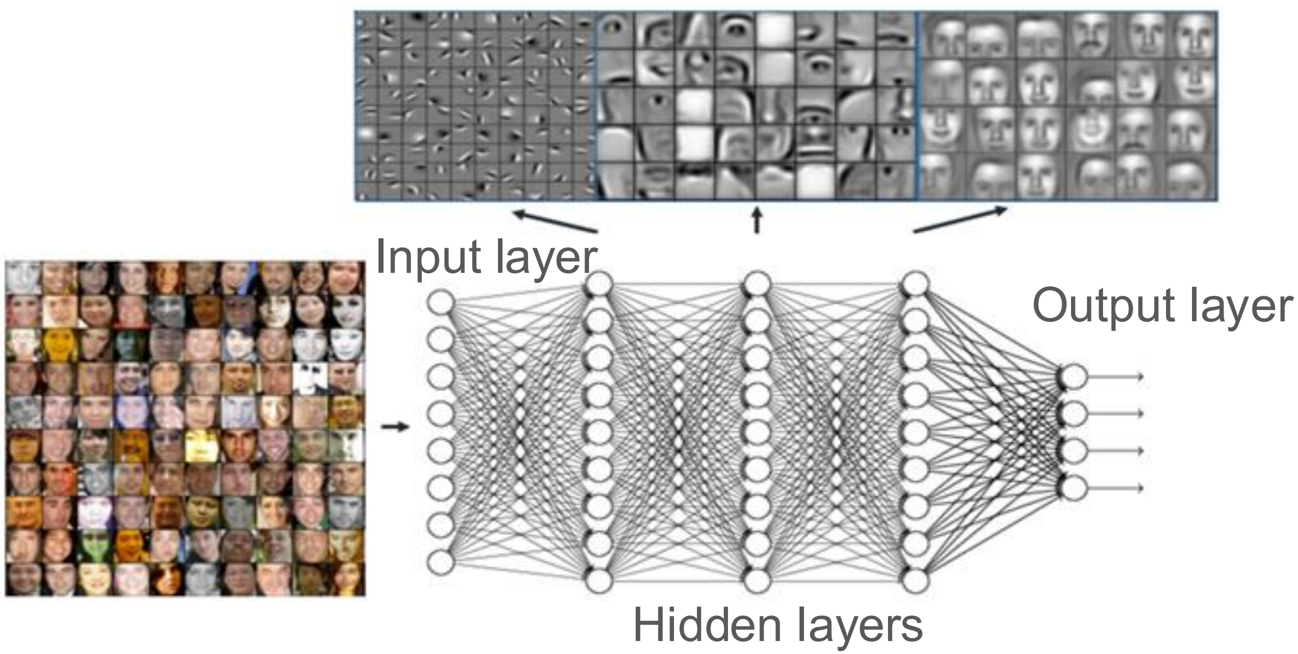
RL: from intuition to artificial systems

Human vs. Agent

[Google's DeepMind AI Just Taught Itself To Walk]()

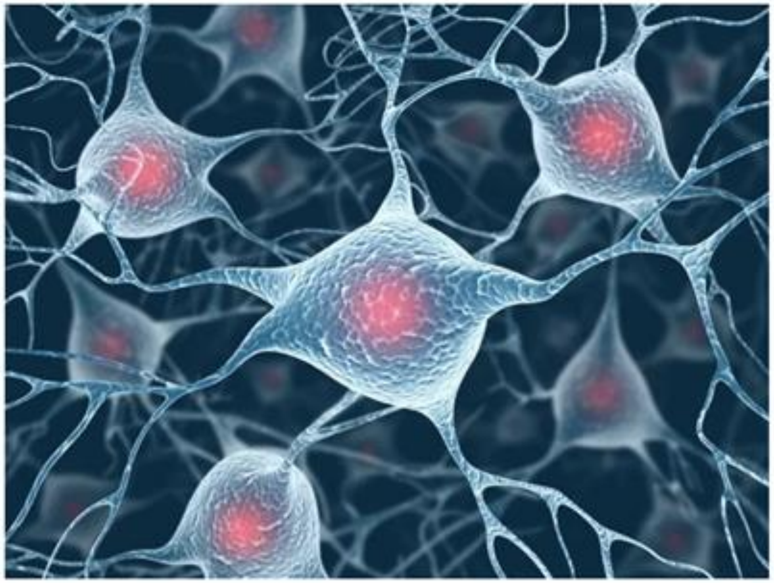
Deep Learning (DL)

- Deep learning uses multi-layer neural networks to automatically learn **representations** from data.
  - Lower layers learn simple features;
  - Higher layers learn abstract concepts;
  - This hierarchy mirrors (in a simplified way) the human nervous system.
- Deep learning underpins modern AI in vision, speech, and language.



DL: from intuition to artificial systems

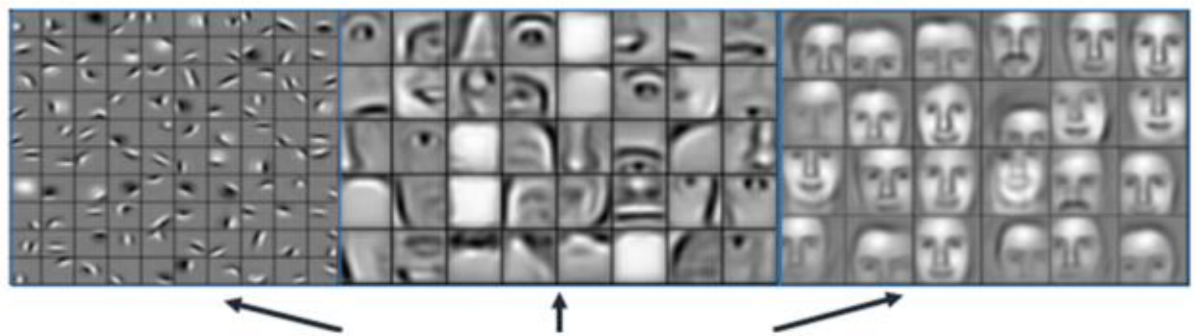
Human nervous system: from neurons to the nervous system

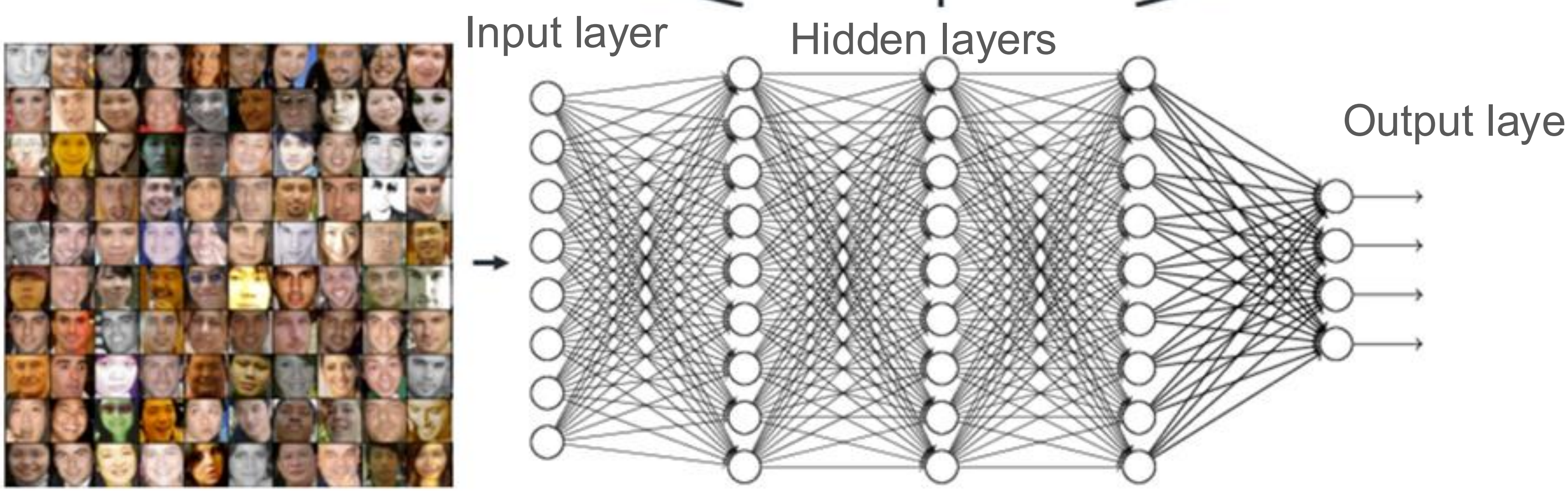


DL: from intuition to artificial systems

Deep learning has **networks** which are capable of learning **representations** from data.

Deep neural networks learn hierarchical feature representations





Deep Reinforcement Learning (DRL)

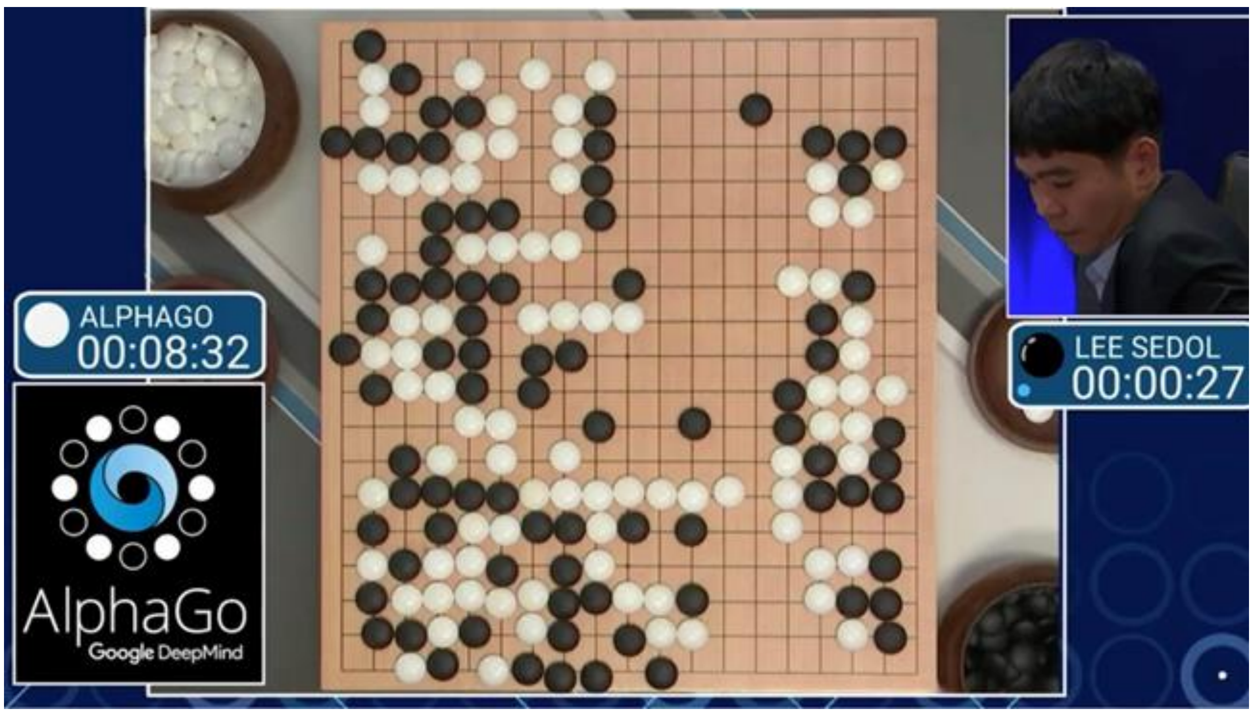
- Deep reinforcement learning combines:
  - RL for sequential decision-making;
  - DL for high-dimensional perception and representation learning.
- Notable successes include AlphaGo and OpenAI Five.
- **Important clarification**: These systems are highly specialised and trained under extensive computational resources. They do not represent general intelligence.

DRL: from intuition to artificial systems

OpenAI Five

DRL: from intuition to artificial systems

[Artificial intelligence: Google's AlphaGo beats Go master Lee Se-dol - BBC News]

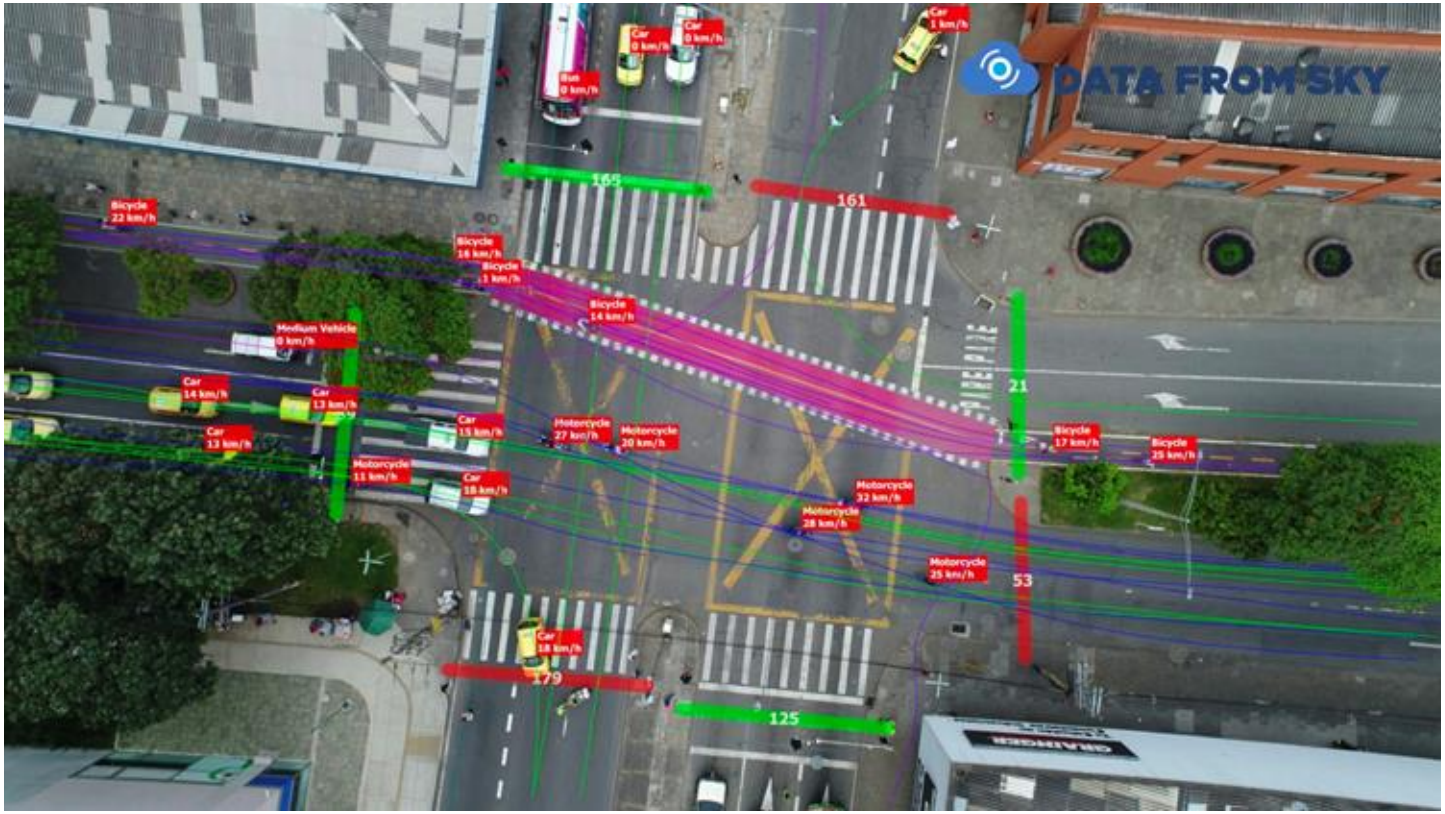


Multi-Agent Systems and Learning

- A **multi-agent system (MAS)** consists of multiple agents interacting within a shared environment.
  - Agents may cooperate or compete;
  - There is no central controller;
  - System-level behaviour is emergent.
- **Multi-agent learning** includes:
  - Multi-agent reinforcement learning;
  - Evolutionary and game-theoretic approaches;
  - Swarm intelligence.
- Applications include traffic control, robotics, and distributed resource management.

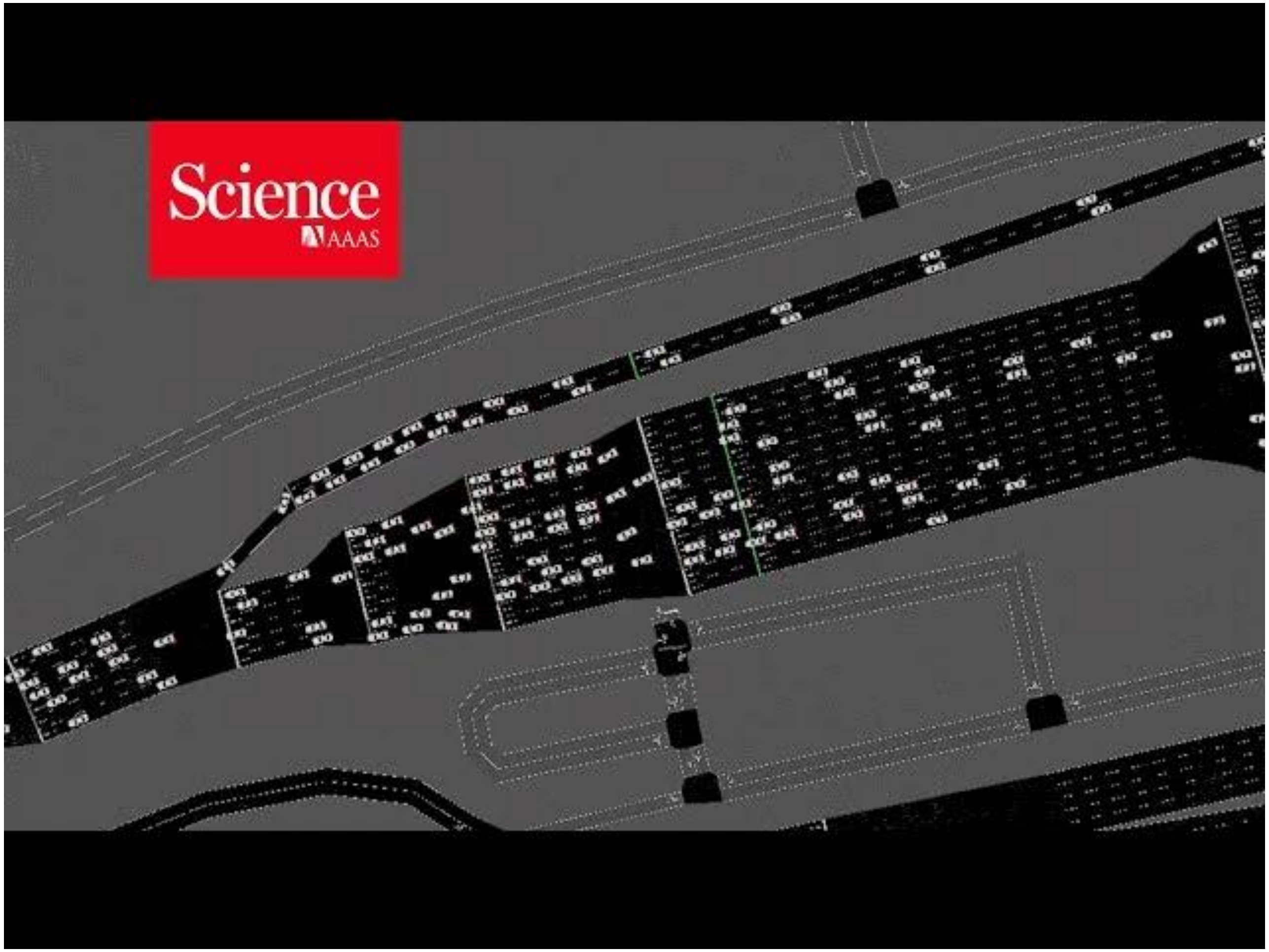
MAL: from intuition to artificial systems

[Traffic analysis from video - multi counter - cars, pedestrians and bicycles counter]



MAL: from intuition to artificial systems

- [AI trained to control traffic]()



Swarm Intelligence

- Swarm intelligence studies how collective intelligence emerges from simple agents following local rules.
- Characteristics:
  - No leader;
  - Local information only;
  - Robust collective behaviour.
- Examples include bird flocks, fish schools, and robot swarms.

Collective ⇒ Individual
No leader
Simple rules ⇒ Intelligence

*Trypophobia Pictures Warning

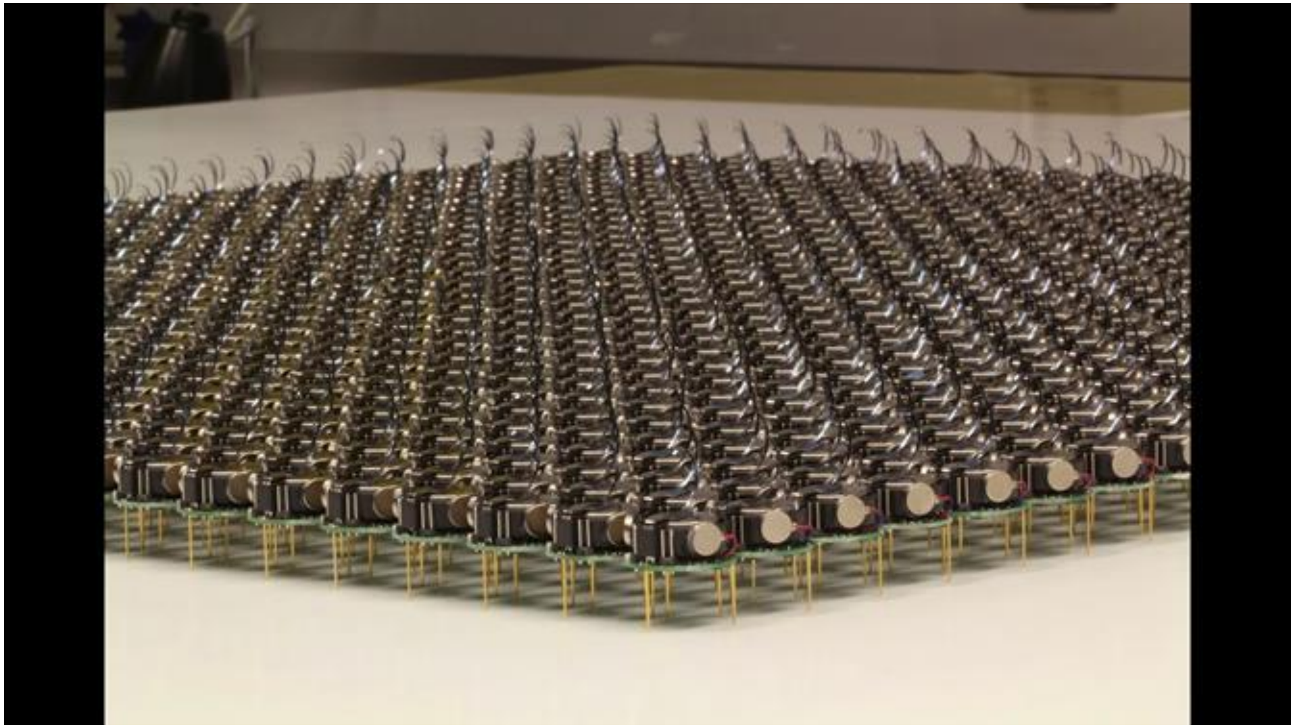
- Please skip the following as it may cause feelings of disgust or fear.

SI: From intuition to artificial systems

[Ten Million Starlings Swarm (7 Tonnes of Bird Poo) | Superswarm | BBC Earth]

SI: From intuition to artificial systems

[A Swarm of One Thousand Robots]()



Summary

- In this lecture you should understand:
  - Why biological systems inspire AI algorithms;
  - The basic ideas of RL, DL, DRL, and MAS;
  - The role of emergence and decentralisation in intelligent systems.